

JavaScript Event Handling Notes

What is an Event?

An **event** is an action performed by the user or browser.

Examples:

- Mouse click
- Mouse movement
- Keyboard press
- Page load
- Form submit

JavaScript listens for these events and performs an action.

1. Change Background Color on Button Click

Event Used

- onclick

Process

1. User clicks the button.
2. JavaScript selects the page (body).
3. Change the backgroundColor property.

Property Used

- style.backgroundColor

Example Colors

- red
 - green
 - blue
 - yellow
 - pink
-

2. Change Heading Color on Mouse Over and Mouse Out

Events Used

onmouseover

Occurs when the mouse pointer enters an element.

Example:

Mouse → Heading

Action:

Heading color becomes **red**.

onmouseout

Occurs when the mouse pointer leaves an element.

Action:

Heading color becomes **black**.

Property Used

style.color

3. Recognize Which Mouse Event is Fired

Mouse events are generated depending on user actions.

Event	Description
onclick	Mouse click
ondblclick	Double click
onmouseover	Mouse enters element
onmouseout	Mouse leaves element
onmousedown	Mouse button pressed
onmouseup	Mouse button released
onmousemove	Mouse moves

The program displays which event occurred.

4. Multiple Mouse Events

(a) Mouse Over Heading

Event

onmouseover

Action

Heading becomes **yellow**.

Event

onmouseout

Action

Heading becomes **black**.

(b) Find Current Date and Time

Event

onclick

Object Used

Date

Method

new Date()

Returns current

- Day
 - Date
 - Month
 - Year
 - Hours
 - Minutes
 - Seconds
-

(c) Red Button

Event

onclick

Action

Background becomes **red**.

(d) Green Button

Event

onclick

Action

Background becomes **green**.

5. Keyboard Events

Keyboard events occur whenever a key is pressed or released.

Keyboard Events

Event	Description
onkeydown	Key is pressed
onkeypress	Character key is pressed (older event)
onkeyup	Key is released

(a) Display Key Code

When a key is pressed,

JavaScript identifies

- Key
- Key code

Examples

Key	Key Code
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A	65
---	----

Enter	13
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Space	32
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Esc	27
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Modern JavaScript prefers **event.key** (e.g., "a", "Enter") over numeric key codes.

(b) Detect Vowel

Compare the pressed key with

- a
- e
- i
- o

- u

If matched

Display

"Vowel key pressed"

Otherwise

"Consonant or other key"

(c) Key Released

Event

onkeyup

Action

Background color changes to **blue**.

6. Days Left Until Next Birthday

Input

User enters

Date of Birth

Steps

Step 1

Read the DOB.

Step 2

Get today's date.

Step 3

Create the birthday date for the current year using:

- Current year
 - Birth month
 - Birth day
-

Step 4

Check

Has birthday already passed this year?

If **No**

Use current year's birthday.

If **Yes**

Use next year's birthday.

Step 5

Subtract

Next Birthday – Today's Date

Result is in **milliseconds**.

Step 6

Convert milliseconds into days.

Formula

Days = Milliseconds ÷ (1000 × 60 × 60 × 24)

Where

1000 = milliseconds in 1 second

60 = seconds in 1 minute

60 = minutes in 1 hour

24 = hours in 1 day

Step 7

Use **Math.ceil()**

Reason

If even a few hours remain,

Example

0.25 day

should become

1 day

instead of

0 day

Step 8

Display

Days left until next birthday: XX

Flow Of Program-6

User selects DOB



Convert DOB to Date object



Get today's date



Create birthday for current year



Has birthday already passed?

Yes | No



Move birthday to next year



Subtract today's date



Difference in milliseconds



Convert milliseconds → days



Display days remaining

Frequently Used DOM Methods

Method	Purpose
getElementById()	Select element using id
getElementsByTagName()	Select elements by tag
getElementsByClassName()	Select elements by class
querySelector()	Select first matching element
querySelectorAll()	Select all matching elements

Frequently Used Style Properties

Property	Purpose
style.color	Change text color
style.backgroundColor	Change background
style.fontSize	Change font size
style.display	Show/Hide element
style.visibility	Make visible/invisible

Common JavaScript Objects

Object	Purpose
Date	Current date and time
Math	Mathematical operations
document	Access HTML page
event	Information about the current event

Important Methods

Method	Use
<code>new Date()</code>	Current date and time
<code>getFullYear()</code>	Current year
<code>getMonth()</code>	Month (0–11)
<code>getDate()</code>	Day of month
<code>getHours()</code>	Hour
<code>getMinutes()</code>	Minutes
<code>getSeconds()</code>	Seconds
<code>Math.ceil()</code>	Round up
<code>Math.floor()</code>	Round down
<code>Math.round()</code>	Round to nearest integer